

Mathematical Properties of the Affine-DDPNM

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All results are stated at the finite-dimensional discrete level. Local-kernel and global-SPD conclusions are conditional on explicit rank and anchoring assumptions. Galerkin optimality is proved relative to a separately defined broken-pressure full-traction-trace system and only in its energy seminorm. Equivalence with the monolithic continuous-pressure Taylor–Hood method is *not* asserted.

1 Discrete setting and sign convention

1.1 Local Stokes discretization

For pore Ω_i , let $A_i \in \mathbb{R}^{n_{u,i} \times n_{u,i}}$ and $B_i \in \mathbb{R}^{n_{p,i} \times n_{u,i}}$ correspond to

$$a_i(\mathbf{u}, \mathbf{v}) = \nu \int_{\Omega_i} \nabla \mathbf{u} : \nabla \mathbf{v} \, dx, \quad b_i(\mathbf{v}, q) = - \int_{\Omega_i} q \nabla \cdot \mathbf{v} \, dx.$$

The local saddle-point system reads

$$K_i \begin{bmatrix} U \\ P \end{bmatrix} := \begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} U \\ P \end{bmatrix} = \begin{bmatrix} f \\ \mathbf{0} \end{bmatrix}. \quad (1)$$

No pressure stabilisation is used.

Assumption 1.1 (Local algebraic stability). *For every pore i :*

- (a) A_i is symmetric positive definite (guaranteed e.g. by a positive-measure no-slip wall segment);
- (b) B_i has full row rank (discrete inf-sup; not implied by (a));
- (c) the coefficient vector $\mathbf{1}_i^P$ of the constant pressure $q_h \equiv 1$ belongs to the local pressure space.

1.2 Interface frames and oriented directions

Each internal interface γ_k is planar and is assigned a global orthonormal frame $\{\mathbf{n}_k, \mathbf{t}_{k,1}, \mathbf{t}_{k,2}\}$ with $\mathbf{n}_k$ directed from the first pore of the pair to the second. Define $\sigma_{i,k} = +1$ if Ω_i is the first member and -1 otherwise. The **oriented inward-traction directions** are

$$\mathbf{d}_{i,k,n} = \mathbf{n}_{i,k} = \sigma_{i,k} \mathbf{n}_k, \quad \mathbf{d}_{i,k,t_\nu} = -\sigma_{i,k} \mathbf{t}_{k,\nu} \quad (\nu = 1, 2). \quad (2)$$

The minus sign in the tangential definition is a pure convention chosen so that the assembled load direction $-\mathbf{d}_{i,k,t_\nu} = +\sigma_{i,k} \mathbf{t}_{k,\nu}$ matches the code's `side_sign * tangent`. Crucially,

$$\mathbf{d}_{j,k,\beta} = -\mathbf{d}_{i,k,\beta} \quad (\beta = n, t_1, t_2). \quad (3)$$

Half-span-normalised in-plane coordinates: $s_k(\mathbf{x}) = ((\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,1})/a_k$, $t_k(\mathbf{x}) = ((\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,2})/b_k$, with $\varphi_0 = 1$, $\varphi_s = s_k$, $\varphi_t = t_k$, $\mathcal{M} = \{0, s, t\} \times \{n, t_1, t_2\}$.

1.3 Traction Ansatz and primitive loads

The unknown coefficients parameterize the **inward** traction:

$$-\boldsymbol{\sigma}(\mathbf{u}_i, p_i) \mathbf{n}_{i,k} = \sum_{(\alpha, \beta) \in \mathcal{M}} \lambda_{k, \alpha, \beta} \varphi_\alpha \mathbf{d}_{i,k, \beta}. \quad (4)$$

The physical outward traction in the weak form is therefore $\boldsymbol{\sigma} \mathbf{n}_{i,k} = -\sum \lambda \varphi \mathbf{d}_{i,k, \beta}$. The primitive load column for mode (α, β) on γ_k is

$$[\mathbf{f}_{i,k}^{(\alpha, \beta)}]_j = - \int_{\gamma_k} \varphi_\alpha \mathbf{d}_{i,k, \beta} \cdot \boldsymbol{\phi}_j^{(i)} dS. \quad (5)$$

The leading minus sign is essential: it converts the inward-traction coefficient into the physical outward-traction boundary integral $\int_{\partial\Omega_i} (\boldsymbol{\sigma} \mathbf{n}) \cdot \mathbf{v} dS$.

Let $L_i \in \mathbb{R}^{n_{u,i} \times n_i}$ collect all primitive columns, $n_i = 9m_i^u + m_i^k$, including one constant-normal column per known boundary port with prescribed coefficient $\lambda_{i,\gamma}^k = p_b$.

1.4 Compliance map and weighted velocity moments

Define

$$M_i = B_i A_i^{-1} B_i^\top, \quad H_i = A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}. \quad (6)$$

Under Assumption 1.1, M_i is SPD.

Lemma 1.2 (Properties of H_i). *H_i is symmetric positive semidefinite and satisfies $H_i^\top A_i H_i = H_i$, $\text{Ker}(H_i) = \text{Range}(B_i^\top)$. For any f , the velocity component of the solution to (1) is $U = H_i f$ and $B_i U = 0$.*

Proof. Let $C_i = A_i^{-1/2} B_i^\top$ and $\Pi_i = C_i (C_i^\top C_i)^{-1} C_i^\top$. Then $M_i = B_i A_i^{-1} B_i^\top = C_i^\top C_i$ is SPD (Assumption 1.1(b) and $A_i \succ 0$). A direct computation gives

$$H_i = A_i^{-1/2} (I - \Pi_i) A_i^{-1/2}.$$

Since $I - \Pi_i$ is an orthogonal projector (symmetric idempotent), H_i is symmetric positive semidefinite, with $\text{Ker}(H_i) = A_i^{1/2} \text{Range}(\Pi_i) = \text{Range}(B_i^\top)$ and $H_i^\top A_i H_i = A_i^{-1/2} (I - \Pi_i) A_i^{-1/2} = H_i$. Solving (1) by block elimination yields $U = H_i f$ and $P = M_i^{-1} B_i A_i^{-1} f$; the second block row then gives $B_i U = B_i H_i f = 0$. $\square$

The local compliance matrix is

$$\mathbf{G}_i = L_i^\top H_i L_i \in \mathbb{R}^{n_i \times n_i}, \quad (7)$$

symmetric positive semidefinite. Partition: $\mathbf{G}_i = \begin{bmatrix} \mathbf{G}_i^{uu} & \mathbf{G}_i^{uk} \\ \mathbf{G}_i^{ku} & \mathbf{G}_i^{kk} \end{bmatrix}$.

The weighted velocity moment for mode (α, β) on interface γ_k of Ω_i is

$$M_{i,k}^{(\alpha, \beta)}(U_i) = \int_{\gamma_k} \varphi_\alpha \mathbf{u}_{i,h} \cdot \mathbf{d}_{i,k, \beta} dS. \quad (8)$$

From the load convention (5), $[\mathbf{G}_i \boldsymbol{\lambda}_i]_{k, \alpha, \beta} = -M_{i,k}^{(\alpha, \beta)}(U_i)$, so $\mathbf{G}_i$ is the *negative* moment response.

The global Schur system, enforcing $M_{i,k}^{(\alpha, \beta)} + M_{j,k}^{(\alpha, \beta)} = 0$ for all nine modes:

$$\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}, \quad \mathbf{S} = \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \quad \mathbf{F} = - \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k. \quad (9)$$

2 Local kernel — conditional characterization

Lemma 2.1 (Symmetry and energy identity). $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = U_i^\top A_i U_i \geq 0$, where $U_i = H_i L_i \boldsymbol{\lambda}_i$.

Proof. $U_i = H_i L_i \boldsymbol{\lambda}_i$ by Lemma 1.2. Then $\boldsymbol{\lambda}_i^\top L_i^\top H_i L_i \boldsymbol{\lambda}_i = (H_i L_i \boldsymbol{\lambda}_i)^\top A_i (H_i L_i \boldsymbol{\lambda}_i)$ using $H_i^\top A_i H_i = H_i$. $\square$

Lemma 2.2 (Exact kernel formula). $\text{Ker}(\mathbf{G}_i) = \text{Ker}(H_i L_i) = \{\boldsymbol{\lambda}_i : L_i \boldsymbol{\lambda}_i \in \text{Range}(B_i^\top)\}$.

Proof. From Lemma 2.1, $\mathbf{G}_i \boldsymbol{\lambda}_i = \mathbf{0}$ iff $H_i L_i \boldsymbol{\lambda}_i = \mathbf{0}$. By Lemma 1.2, $H_i f = 0$ iff $f \in \text{Range}(B_i^\top)$. Hence $H_i L_i \boldsymbol{\lambda}_i = 0$ iff $L_i \boldsymbol{\lambda}_i \in \text{Range}(B_i^\top)$. Note: no claim that H_i is SPD on $\text{Range}(L_i)$ is made or needed; that claim would be false because $\text{Range}(L_i) \ni B_i^\top \mathbf{1}_i^p \in \text{Ker}(H_i)$. $\square$

Let $\mathbf{1}_{i,0n} \in \mathbb{R}^{n_i}$ have entry 1 at every constant-normal mode $(\gamma, 0, n)$ (internal and boundary) and 0 elsewhere.

Lemma 2.3 (Constant-pressure null vector). $L_i \mathbf{1}_{i,0n} = B_i^\top \mathbf{1}_i^p$. Hence $\mathbf{1}_{i,0n} \in \text{Ker}(\mathbf{G}_i)$.

Proof. Summing all constant-normal port columns: $[L_i \mathbf{1}_{i,0n}]_j = -\int_{\partial\Omega_i} \phi_j^{(i)} \cdot \mathbf{n}_i \, dS = -\int_{\Omega_i} \nabla \cdot \phi_j^{(i)} \, dx = [B_i^\top \mathbf{1}_i^p]_j$. Then Lemma 2.2 applies. $\square$

Assumption 2.4 (No non-constant pressure-load dependency). For each pore,

$$L_i \boldsymbol{\lambda}_i \in \text{Range}(B_i^\top) \implies \boldsymbol{\lambda}_i = c \mathbf{1}_{i,0n} \text{ for some } c \in \mathbb{R}. \quad (10)$$

Equivalently, $\text{rank}(H_i L_i) = n_i - 1$.

Theorem 2.5 (Conditional local-kernel characterization). Under Assumptions 1.1 and 2.4, $\text{Ker}(\mathbf{G}_i) = \text{span}\{\mathbf{1}_{i,0n}\}$.

Proof. Lemma 2.3 gives $\supseteq$. For $\subseteq$, any $\boldsymbol{\lambda}_i \in \text{Ker}(\mathbf{G}_i)$ satisfies $L_i \boldsymbol{\lambda}_i \in \text{Range}(B_i^\top)$ by Lemma 2.2. Assumption 2.4 forces $\boldsymbol{\lambda}_i = c \mathbf{1}_{i,0n}$. $\square$

Remark 2.6 (Interpretation of the rank condition). Equation (10) states that the n_i responses $H_i \mathbf{f}_i^{(\mu)}$ satisfy exactly one linear dependency—the constant-pressure relation of Lemma 2.3. It does not assert that these responses have codimension 1 in $\text{Range}(H_i)$; that codimension is $\text{rank}(H_i) - (n_i - 1)$ and is generally large. Per-face linear independence of the nine scalar/vector functions is necessary but not sufficient for Assumption 2.4: it does not exclude cross-interface dependencies or dependencies introduced by the incompressible projection H_i .

3 Global Schur complement — SPD and mass conservation

For pore i , let $J_i^u : \mathbb{R}^{9m_i^u} \rightarrow \mathbb{R}^{n_i}$ be the zero extension into the known boundary slots: $J_i^u(z) = (z, \mathbf{0})$. The global energy is

$$\boldsymbol{\lambda}^\top \mathbf{S} \boldsymbol{\lambda} = \sum_{i=1}^N (J_i^u R_i^u \boldsymbol{\lambda})^\top \mathbf{G}_i (J_i^u R_i^u \boldsymbol{\lambda}). \quad (11)$$

Assumption 3.1 (Graph anchoring). Every connected component of the pore adjacency graph contains at least one pore with at least one prescribed-traction (inlet/outlet) port.

Theorem 3.2 (Global SPD and unique solvability). Under Assumptions 1.1, 2.4, and 3.1, $\mathbf{S}$ is SPD.

Proof. Symmetry and positive semidefiniteness follow from (11) and Lemma 2.1. Suppose $\boldsymbol{\lambda}^\top \mathbf{S} \boldsymbol{\lambda} = 0$. Every summand vanishes, so Theorem 2.5 gives $J_i^u R_i^u \boldsymbol{\lambda} = c_i \mathbf{1}_{i,0n}$ for each pore.

If pore i has a boundary port, its slot in $J_i^u R_i^u \boldsymbol{\lambda}$ is zero (by the zero extension) but the corresponding slot in $\mathbf{1}_{i,0n}$ is 1; hence $c_i = 0$.

For adjacent pores i, j sharing an interface, the same global coefficient appears in both restrictions $R_i^u \boldsymbol{\lambda}$ and $R_j^u \boldsymbol{\lambda}$, forcing $c_i = c_j$. The constants are thus uniform on each connected graph component. Assumption 3.1 ensures every component contains a boundary pore where $c = 0$; hence all $c_i = 0$ and $\boldsymbol{\lambda} = \mathbf{0}$. $\square$

Remark 3.3 (Floating basins are distinct from anchoring). **Local coercivity** (Assumption 1.1(a)) fails for pores without a solid-wall facet (floating basins); these are merged into neighbours before assembly. **Graph anchoring** (Assumption 3.1) is a separate condition requiring an inlet/outlet port in each connected component. Merging a floating basin restores coercivity but does not supply graph anchoring.

Theorem 3.4 (Discrete mass conservation). *Under Assumption 1.1, each local reconstructed velocity satisfies $\sum_k M_{i,k}^{(0,n)} = 0$. If, in addition, the global Schur system (9) admits a solution $\boldsymbol{\lambda}$, then the assembled velocity field satisfies $\int_{\partial\Omega} \mathbf{u}_h^{DD} \cdot \mathbf{n} \, dS = 0$.*

Proof. $B_i U_i = 0$ and the constant test function $\mathbf{1}_i^p$ give the local balance. The Schur equations enforce $M_{i,k}^{(0,n)} + M_{j,k}^{(0,n)} = 0$; summing over pores cancels internal fluxes, leaving only the boundary contributions which sum to zero. $\square$

A Error estimate as a Galerkin projection

We define a broken-pressure full-traction-trace (C-DD) reference system and prove that the Affine-DDPNM is its Galerkin restriction. This reference system uses *discontinuous* pressure across pores ($Q_h^{\text{br}} = \bigoplus_i Q_{i,h}$); it is not the monolithic continuous- P_1 Taylor–Hood system.

A.1 Full-traction-trace (C-DD) system

On interface γ_k , let $\{\psi_{k,\ell}\}_{\ell=1}^{m_k}$ be a scalar trace basis (the restriction of the $[P_2]^3$ velocity trace). The C-DD variable $\boldsymbol{\xi}$ is defined to represent the **inward** traction $-\boldsymbol{\sigma} \mathbf{n}$ at mesh nodes, matching the affine convention (4). Define coupling blocks using the same oriented directions as in (2):

$$[N_{i,k}^\beta]_{\ell,j} = \int_{\gamma_k} \psi_{k,\ell} \mathbf{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)} \, dS, \quad \beta \in \{n, t_1, t_2\}. \quad (12)$$

For an inward-traction nodal vector $\boldsymbol{\xi}_{i,k} = (\xi^n; \xi^{t_1}; \xi^{t_2}) \in \mathbb{R}^{3m_k}$, the physical outward traction is $\boldsymbol{\sigma} \mathbf{n} = -\boldsymbol{\xi}$; the weak-form boundary load is therefore $-N_i^\top \boldsymbol{\xi}_i$, matching the load convention (5).

The local C-DD Schur complement is $\widehat{\mathbf{S}}_i = N_i H_i N_i^\top$. Boolean assembly uses standard maps $\widehat{R}_i^u$ (no additional sign factor needed: the orientation $\mathbf{d}_{j,k,\beta} = -\mathbf{d}_{i,k,\beta}$ from (3) is already encoded in the coupling blocks $N_{i,k}^\beta$ and the globally shared coefficient blocks). Assembly yields

$$\widehat{\mathbf{S}} \boldsymbol{\xi}^{\text{full}} = \widehat{\mathbf{F}}, \quad \widehat{\mathbf{S}} = \sum_i (\widehat{R}_i^u)^\top \widehat{\mathbf{S}}_i^{uu} \widehat{R}_i^u. \quad (13)$$

$\widehat{\mathbf{S}}$ is symmetric positive semidefinite. Define $\|x\|_{\widehat{\mathbf{S}}} = (x^\top \widehat{\mathbf{S}} x)^{1/2}$; without a full-trace kernel theorem this is a seminorm.

Let $m_{\Gamma,i}^k := 3 \sum_{\gamma \in \mathcal{K}_i} m_{i,\gamma}$ be the total number of C-DD boundary trace DOFs on Ω_i , where $m_{i,\gamma}$ is the number of scalar trace nodes on boundary port γ . The C-DD boundary data vector is $\boldsymbol{\xi}_i^k \in \mathbb{R}^{m_{\Gamma,i}^k}$, collecting the known inward-traction nodal values on all boundary ports of Ω_i .

The corresponding coupling blocks $N_i^k \in \mathbb{R}^{m_{\Gamma,i}^k \times n_{u,i}}$ are assembled from (12) restricted to the boundary ports, and the C-DD right-hand side is

$$\widehat{\mathbf{F}} = - \sum_{i=1}^N (\widehat{R}_i^u)^\top N_i^u H_i (N_i^k)^\top \boldsymbol{\xi}_i^k. \quad (14)$$

In the affine system, each boundary port carries only the single constant-normal mode. We define the boundary restriction map $P_i^k \in \mathbb{R}^{m_{\Gamma,i}^k \times m_i^k}$ columnwise: each boundary port $\gamma \in \mathcal{K}_i$ contributes one column, the L^2 -projection of $\varphi_0 = 1$ onto the scalar trace basis on that port, placed in the normal block-row. The boundary nodal inward traction is then $\boldsymbol{\xi}_i^k = P_i^k R_i^k \boldsymbol{\lambda}^k$, and the load identity extends to $L_i^k = -(N_i^k)^\top P_i^k$.

Assumption A.1 (Matching interface trace spaces). *For every internal interface γ_k shared by pores i and j , the two local velocity trace spaces coincide: $\mathbf{V}_{i,h}|_{\gamma_k} = \mathbf{V}_{j,h}|_{\gamma_k}$, with a shared scalar trace basis $\{\psi_{k,\ell}\}$ and a one-to-one DOF correspondence between the two sides. The Boolean assembly maps $\widehat{R}_i^u$ operate on this shared nodal basis.*

A.2 Affine restriction and algebraic equivalence

Let M_k be the face mass matrix and $p_k^\alpha = M_k^{-1} b_k^\alpha$, $(b_k^\alpha)_\ell = \int_{\gamma_k} \varphi_\alpha \psi_{k,\ell} dS$, $\alpha \in \{0, s, t\}$. For the unconstrained full $[P_2]^3$ trace space on a triangular face, $\{\varphi_0, \varphi_s, \varphi_t\}$ are exactly representable; the L^2 -projection then satisfies $M_k p_k^\alpha = b_k^\alpha$ with pointwise equality. If trace DOFs are eliminated by no-slip wall constraints, only the mass-matrix orthogonality is guaranteed; the proof below uses only this orthogonality and does not rely on pointwise equality.

With coefficient order $(0, n), (0, t_1), (0, t_2), (s, n), (s, t_1), (s, t_2), (t, n), (t, t_1), (t, t_2)$, the correct restriction matrix is

$$P_k^{\text{aff}} = \begin{bmatrix} p_k^0 & 0 & 0 & p_k^s & 0 & 0 & p_k^t & 0 & 0 \\ 0 & p_k^0 & 0 & 0 & p_k^s & 0 & 0 & p_k^t & 0 \\ 0 & 0 & p_k^0 & 0 & 0 & p_k^s & 0 & 0 & p_k^t \end{bmatrix} \in \mathbb{R}^{3m_k \times 9}. \quad (15)$$

Each column occupies only the block-row matching its direction β . The global restriction is $\mathcal{P}_{\text{aff}} = \bigoplus_k P_k^{\text{aff}} : \mathbb{R}^{9m} \rightarrow \mathbb{R}^{m_\Gamma}$, where $m_\Gamma = 3 \sum_{k=1}^m m_k$ is the total number of C-DD interface traction DOFs (three components per trace node per interface).

Lemma A.2 (Algebraic equivalence). *Under Assumption A.1, $\mathbf{S} = \mathcal{P}_{\text{aff}}^\top \widehat{\mathbf{S}} \mathcal{P}_{\text{aff}}$, $\mathbf{F} = \mathcal{P}_{\text{aff}}^\top \widehat{\mathbf{F}}$.*

Proof. From (5), (12), and (15), $L_i^u = -(N_i^u)^\top P_i^{\text{aff}}$ (the minus sign comes from the load convention). Consequently $\mathbf{G}_i^{uu} = (L_i^u)^\top H_i L_i^u = (P_i^{\text{aff}})^\top N_i^u H_i (N_i^u)^\top P_i^{\text{aff}}$ (the minus signs cancel quadratically). Summing with the Boolean maps and using the commutativity of the block-diagonal restriction with the global interface indexing gives $\mathbf{S} = \mathcal{P}_{\text{aff}}^\top \widehat{\mathbf{S}} \mathcal{P}_{\text{aff}}$. For the forcing term: the boundary counterpart of the load identity is $L_i^k = -(N_i^k)^\top P_i^k$. From (14) and $\boldsymbol{\xi}_i^k = P_i^k R_i^k \boldsymbol{\lambda}^k$,

$$\begin{aligned} \mathcal{P}_{\text{aff}}^\top \widehat{\mathbf{F}} &= \sum_i (R_i^u)^\top (P_i^{\text{aff}})^\top (-N_i^u H_i (N_i^k)^\top P_i^k R_i^k \boldsymbol{\lambda}^k) \\ &= - \sum_i (R_i^u)^\top (-(L_i^u)^\top) H_i (-L_i^k) R_i^k \boldsymbol{\lambda}^k \\ &= - \sum_i (R_i^u)^\top (L_i^u)^\top H_i L_i^k R_i^k \boldsymbol{\lambda}^k = - \sum_i (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k = \mathbf{F}, \end{aligned}$$

where we used $L_i^u = -(N_i^u)^\top P_i^{\text{aff}}$, $L_i^k = -(N_i^k)^\top P_i^k$, and the definition $\mathbf{G}_i^{uk} = (L_i^u)^\top H_i L_i^k$. The three minus signs from the load identities and the C-DD right-hand side produce a net single minus sign, exactly matching the definition of $\mathbf{F}$ in (9). $\square$

A.3 Error estimate and enrichment

Theorem A.3 (Best approximation in the full-trace energy seminorm). *Under Assumption A.1, assume $\widehat{\mathbf{S}} \boldsymbol{\xi}^{\text{full}} = \widehat{\mathbf{F}}$ is consistent. Let $\boldsymbol{\lambda}^{\text{aff}}$ solve $\mathbf{S} \boldsymbol{\lambda}^{\text{aff}} = \mathbf{F}$ and set $\boldsymbol{\xi}^{\text{aff}} = \mathcal{P}_{\text{aff}} \boldsymbol{\lambda}^{\text{aff}}$ (both represent inward traction $-\boldsymbol{\sigma} \mathbf{n}$). Then*

$$\|\boldsymbol{\xi}^{\text{full}} - \boldsymbol{\xi}^{\text{aff}}\|_{\widehat{\mathbf{S}}} = \min_{\boldsymbol{\mu} \in \mathbb{R}^{9m}} \|\boldsymbol{\xi}^{\text{full}} - \mathcal{P}_{\text{aff}} \boldsymbol{\mu}\|_{\widehat{\mathbf{S}}}, \quad (16)$$

$$\sum_i \|U_i^{\text{full}} - U_i^{\text{aff}}\|_{A_i}^2 = \|\boldsymbol{\xi}^{\text{full}} - \boldsymbol{\xi}^{\text{aff}}\|_{\widehat{\mathbf{S}}}^2. \quad (17)$$

The statements remain valid when $\widehat{\mathbf{S}}$ is only semidefinite; minimizers need not be unique but seminorm values are well defined.

Proof. Lemma A.2 reduces the Affine system to the Galerkin restriction. Subtracting $\mathcal{P}_{\text{aff}}^\top (\widehat{\mathbf{S}} \boldsymbol{\xi}^{\text{full}} - \widehat{\mathbf{F}}) = \mathbf{0}$ from $\mathbf{S} \boldsymbol{\lambda}^{\text{aff}} - \mathbf{F} = \mathbf{0}$ gives $\mathcal{P}_{\text{aff}}^\top \widehat{\mathbf{S}} (\boldsymbol{\xi}^{\text{aff}} - \boldsymbol{\xi}^{\text{full}}) = \mathbf{0}$. Hence the error $\mathbf{e} = \boldsymbol{\xi}^{\text{full}} - \boldsymbol{\xi}^{\text{aff}}$ is $\widehat{\mathbf{S}}$ -orthogonal to $\text{Range}(\mathcal{P}_{\text{aff}})$. For any $\boldsymbol{\mu}$, let $\mathbf{v} = \boldsymbol{\xi}^{\text{aff}} - \mathcal{P}_{\text{aff}} \boldsymbol{\mu}$. The cross term vanishes, so $\|\mathbf{e} + \mathbf{v}\|_{\widehat{\mathbf{S}}}^2 = \|\mathbf{e}\|_{\widehat{\mathbf{S}}}^2 + \|\mathbf{v}\|_{\widehat{\mathbf{S}}}^2 \geq \|\mathbf{e}\|_{\widehat{\mathbf{S}}}^2$. Equality at $\boldsymbol{\mu} = \boldsymbol{\lambda}^{\text{aff}}$ proves (16).

For (17), let $\delta \boldsymbol{\xi} = \boldsymbol{\xi}^{\text{full}} - \boldsymbol{\xi}^{\text{aff}}$. The local velocity difference is $\delta U_i = -H_i (N_i^u)^\top \widehat{R}_i^u \delta \boldsymbol{\xi}$ (the minus sign matches the load convention (5)). Using $H_i^\top A_i H_i = H_i$ from Lemma 1.2,

$$\begin{aligned} \sum_i \|\delta U_i\|_{A_i}^2 &= \sum_i (\delta U_i)^\top A_i (\delta U_i) \\ &= \sum_i (\widehat{R}_i^u \delta \boldsymbol{\xi})^\top N_i^u H_i (N_i^u)^\top (\widehat{R}_i^u \delta \boldsymbol{\xi}) = \delta \boldsymbol{\xi}^\top \widehat{\mathbf{S}} \delta \boldsymbol{\xi}, \end{aligned}$$

which is exactly $\|\boldsymbol{\xi}^{\text{full}} - \boldsymbol{\xi}^{\text{aff}}\|_{\widehat{\mathbf{S}}}^2$. $\square$

Corollary A.4 (Affine versus classic enrichment). *Let $\mathcal{P}_{P0} : \mathbb{R}^m \rightarrow \mathbb{R}^{m_r}$ be the restriction to constant normal modes only (each interface contributes one column: the L^2 -projection of $\varphi_0 = 1$ onto the trace basis, in the normal block-row only). Let $\boldsymbol{\lambda}^{P0}$ solve the classic Galerkin system $(\mathcal{P}_{P0}^\top \widehat{\mathbf{S}} \mathcal{P}_{P0}) \boldsymbol{\lambda}^{P0} = \mathcal{P}_{P0}^\top \widehat{\mathbf{F}}$ and set $\boldsymbol{\xi}^{P0} = \mathcal{P}_{P0} \boldsymbol{\lambda}^{P0}$. Since $\text{Range}(\mathcal{P}_{P0}) \subset \text{Range}(\mathcal{P}_{\text{aff}})$,*

$$\|\boldsymbol{\xi}^{\text{full}} - \boldsymbol{\xi}^{\text{aff}}\|_{\widehat{\mathbf{S}}} \leq \|\boldsymbol{\xi}^{\text{full}} - \boldsymbol{\xi}^{P0}\|_{\widehat{\mathbf{S}}}. \quad (18)$$

No analogous monotonicity follows for velocity L^2 , pressure L^2 , broken H^1 , or outlet flux errors.

B Statements that are *not* established

- Unconditional local rank.** Assumption 2.4 is not proved for general Lipschitz pores or for all meshes. Numerical observation of a single near-zero eigenvalue of $\mathbf{G}_i$ for tested geometries is evidence, not a general theorem.
- Per-face independence implies SPD.** This implication is false as a proof principle. The valid global result is Theorem 3.2, conditional on Assumption 2.4.
- Broken-pressure C-DD equals monolithic FEM.** The full-trace system uses independent local pressure spaces $Q_h^{\text{br}} = \bigoplus_i Q_{i,h}$, while the monolithic Taylor–Hood problem uses globally continuous Q_h^c . The implemented “Exact FE-trace Schur” is a primal Schur reduction of the monolithic system and verifies its own block-elimination algebra, not the equivalence with the broken-pressure dual system used in Theorem A.3.
- Continuum or mesh-independent convergence.** All results are finite-dimensional and mesh-dependent. Continuum limits require stable trace-space limits and a fixed interface topology. Watershed partitions whose basin count varies with mesh size do not satisfy the latter.

C Practical algebraic diagnostics

For each pore, the following checks directly test the hypotheses:

1. Verify symmetry and positive definiteness of A_i ;
2. Verify the numerical row rank of B_i ;
3. Verify $\|L_i \mathbf{1}_{i,0n} - B_i^\top \mathbf{1}_i^p\|$ at assembly tolerance (Lemma 2.3);
4. Compute singular values of $H_i L_i$ and confirm that one singular value is numerically zero (within a stated tolerance, e.g. 10^{-12} relative to the largest), with the corresponding right singular vector aligned with $\mathbf{1}_{i,0n}$ (check the angle or normalized residual, not just the singular value). As a necessary condition, verify the dimension bound $n_i - 1 \leq \text{rank}(H_i) = n_{u,i} - n_{p,i}$; if this fails, Assumption 2.4 cannot hold.
5. After global assembly, verify symmetry and positive definiteness of $\mathbf{S}$ (do not infer solely from geometry);
6. Verify local mass, interface moment, and global boundary-flux residuals independently.

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